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Impacts of energy efficiency policies on the integration of renewable energy

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A R T I C L E I N F O

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A B S T R A C T

Energy transitions in developing countries principally consist of the integration of renewable energies and energy efficiency in order to achieve a sustainability framework. Particularly, energy stakeholders are responsible for power planning to assure a country transition. Alongside, they are responsible for providing access to energy through system reliability. The objective of this paper is to demonstrate how a planned decrease of power system reliability, without impacting access to energy, could lead to a better integration of renewable energies. Therefore, we have linked an energy efficiency action, the peak clipping as scheduled outages, to the power system reliability factor. This factor has been integrated in the energy planning model, Open Source energy Modeling SYStem (OSeMOSYS) which is a cost-based long-term optimization model. Following, enhanced OSeMOSYS has been applied to the Tunisia power system using two scenarios reflecting renewable energy sources penetration. Simulation results showed a high rate of renewable energy sources penetration with a decrease of the power system reliability relying on energy efficiency actions.

1. Introduction

Facing the alarming situation of global warming and fossil fuels shortage in several developing countries, Governments have opted to energy transitions, through changes in the system structure, economics and policies, in the 21st century (Débat national sur l'énergie), (British Petroleum, 2019). Transitions towards sustainable energy have been primarily driven by the integration of renewable energies (RE), energy efficiency (EE) actions and clean technologies such as electric vehicles. Among the objectives of energy efficiency, we can cite the rational use of energy, the decrease of energy intensity and the demand. Meanwhile, the satisfaction of the energy demand and ensuring energy access are priorities for Governments (Boisgibault, 2015), (Carafa et al., 2016).

In fact, electrification represents an instrument to quantify energy access through the power system development. The satisfaction of electricity demand is reached by i) generating capacities' investments planning and ii) dispatching. While i) targets the satisfaction of electricity demand on a medium and long-term perspective, ii) satisfies the instantaneous behavior of the demand. Both are reflected through the concept of system reliability, i.e. the rate of the satisfaction of the demand of all customers. Since the 1930's, system reliability has been the subject of several research studies that have defined its hierarchical levels and its interdependency with economics and assessment tools (Allan, 1994), (Vrana and Johansson, 2011). There exists two time frames of reliability: long-terms (hours and years) and short-terms (seconds to minutes). The first type, discussed in this paper, is managed by the system operator and decision makers through the development and use of operating and capacity reserves. Planning generating capacities' investments includes fossil fuel, RE and storage technologies. The second type of reliability is managed by the system operator during dispatching.

Achieving and maintaining adequate system reliability has evolved alongside changes in the power system. In the past, conventional power plants were used to satisfy demand. Also, natural gas turbines, seen as a flexibility tool, were appealed for ramping and peak demand satisfaction. And spinning reserves were a source of stabilizing frequency and controlling voltage. Nowadays with the high rates of RES penetration subject to intermittency, novel technologies, such as smart grid and electric vehicles, are used. These technologies respond to the needs of the system operator especially during the peak and the trough to smoothen the load curve and equilibrate the production and the demand. Furthermore, a new generation of gas turbines are developed and integrated to assure meeting the basic load as well as fast ramping,

Abbreviations: CCGTs, Combined Cycle Gas Turbines; CSP, Concentrated Solar Power plant; EE, Energy Efficiency; OCGTs, Open Cycle Gas Turbines; OSeMOSYS, Open Source energy Modeling SYStem; PPs, Power Plants; PSH, Pumped-Storage Hydroelectricity; RE, Renewable Energies; RES, Renewable Energy Sources

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for example combined cycle power plants could achieve up to 20MW/min) (U.S. Department of Energy, 2016).

In addition, among EE actions, load management is demand-oriented and tries to adjust the load curves so the demand during the peak period is decreased. Different methods are applied, such as peak clipping, valley lifting, load shifting, etc. (Bhattacharyya, 2011a; Medjoudj et al., 2017; Prada, 1999; Lannoye, 2015; Jones, 2017). Peak clipping consists of reducing the demand during peak periods through scheduled outages. However, it exists another method of decreasing the peak, representing a mediocre solution impacting system reliability, which is load shedding. It consists of cutting electricity supply by zone during a period to manage the demand and avoid blackouts. Even though both approaches aim to reduce the demand during the peak; peak clipping is seen as an EE approach and load shedding reflects an inefficient load management approach. This paper argues that scheduled decrease of system reliability could be seen as an EE action. Several questions arise about system reliability decrease, especially that system operators and decision makers target achieving 100% of system reliability (Kjølle, 2017).

In this paper, we present insights about system reliability impact on RE integration. Indeed, system reliability is associated to actions of EE, i.e. decreasing the peak load. In section 2, we present the choice of the tool Open Source energy Modeling SYStem (OSeMOSYS) to model energy systems. We introduce the relationship of system reliability and EE actions followed by system reliability integration in OSeMOSYS. We present needed data for the application of the adopted methodology on the power system of Tunisia as a case study. Following in section 3, we apply the integrated system reliability in the core code of OSeMOSYS to the case of Tunisia power system. We simulate and compare two RE supply-based scenarios, i.e. the Business As Usual (BAU) and RE scenarios taking into consideration two principal decision variables: the future needed investments to satisfy electricity demand and costs. This comparison will allow pointing out the importance of considering EE actions alongside the objective of integrating RE in the power system.

2. Methodology and data

This section argues the choice of the tool to model energy systems, describes the main characteristics of the model and introduces the concept of system reliability in energy system modeling. Furthermore, it includes the power system of Tunisia and RE national objective as a case study to present insights on system reliability impacts of RE integration. The objective of achieving a certain rate of penetration in terms of RE is pointed out in OSeMOSYS using a supply-based scenarios comparison; where scenarios reflect the shares of conventional and renewable energies.

2.1. Tool: Open Source energy Modeling SYStem

The developed countries have been using Energy Systems Modeling (ESM) (Kern and Smith, 2008) since the 1950's for energy transitions. This has led to a high number of approaches and types. The different types of analyses which were primarily developed target the supply side of an energy system, where the objective was to meet a given exogenous energy demand (Bhattacharyya, 2011b). Due to the oil crisis in the 1970's, demand side energy systems analyses have appeared and have kept evolving through years, leading to forecasting approaches, and later on translated to top-down models. Meanwhile, supply-oriented focus has kept evolving and has led to integrated bottom-up models which are technologically oriented to identify the needed investments or to operate short-term solutions (Forum et al., 1977; Hoffman and Jorgenson, 1977; Hoffman and Wood, 1976). Lastly, a combination of bottom-up and top-down models has resulted in better insights for decision makers (Böhringer, 1998; Ortiz and Markandya, 2010; Gargiulo and Gallachóir, 2013; Bhattacharyya and Timilsina, 2009). In brief, the various developed models aim in general to introduce a better energy supply system design with an improved understanding of the present and future interactions between demand-supply, environment, and economy (Hoffman and Wood, 1976). Considering the review articles on ESM (Bhattacharyya and Timilsina, 2010; Pandey, 2002; Urban et al., 2007), the choice of the appropriate modeling framework is identified from the context and the challenges depicted in major developing countries. In fact, it has been proven that the problem of optimally designing the electricity system, in developing countries, in such a way as to match the demand with available resources and supply technologies can be solved by a bottom-up optimization modeling framework. Given the easy access to the source code, the different developed versions, and the accessibility to manuals, we have chosen OSeMOSYS to model the power system and integrate reliability factors (Howells et al., 2011), (Bazilianet al., 2012). In (Howells et al., 2011), developers have introduced OSeMOSYS as an open-source energy system model with a medium-to long-term time horizon. OSeMOSYS is a bottom-up, dynamic and linear optimization model applied to the integrated assessment and energy planning.

Different versions of OSeMOSYS have been developed since its creation in 2011 to improve simulation conditions (for e.g. timing and relaxing optimization), add coding blocks for energy systems (such as storage, short-term flexibility, interconnections, etc.) and better model reality.

OSeMOSYS aims to determine the lowest net present cost of an energy system to meet given demands and constraints. The objective function of OSeMOSYS consists of minimizing the total discounted cost of energy production. Equation (1) represents the constraint of demand satisfaction.

$$Min\ \ total\ \ NPV\ \ cost = Min \sum_{r,y} Total\ \ discounted\ \ cost_{r,y} \tag{1}$$

Where:

- r: Region – country modeled
- y: Year in time horizon
- *Total NPV cost* is the system total net present value cost
- *Total discounted cost* is the system total discounted cost including fixed costs and variable costs at a certain discount rate

The total cost of energy production considers the costs of investment, the variable and fixed costs of operations and maintenance, financial penalties on emissions and residual capacities financial values.

The structure of OSeMOSYS model is developed in a series of component ‘blocks’ of functionality (Howells et al., 2011). These blocks are related to costs, capacity adequacies (i.e. needed installed generating capacities), energy balance (i.e. the satisfaction of energy demand), renewable energies (i.e. the achievement of a certain rate of RE penetration), emissions (i.e. the respect of emissions limits and application of penalties) and reserves (i.e. the respect of reserve margin and satisfaction of primary and secondary reserves). Each block is characterized by parameters, added by the analyst, intermediate variables, equations and constraints. Each block also has a principle objective revealed through a constraint and is composed of a plain English description of sets, parameters, variables, constraints and objectives. The plain English description helps on matching policies and insights with energy system analyst's expectations (Howells et al., 2011).

For example, equation (2) is intended to meet the demand of energy:

$$\forall_{r,l,f,y}\ production_{r,l,f,y} \geq demand_{r,l,f,y} + use_{r,l,f,y} \tag{2}$$

Where:

- r: Region – country modeled
- l: Time slice – fraction representing a period in a year
- y: Year in time horizon
- f: 'Fuel' used or generated by a technology
- $Production_{r,l,f,y}$ is the production of a certain fuel, i.e. energy, to satisfy the demand for each region, time slice, fuel and year
- $Demand_{r,l,f,y}$ is the demanded fuels for each region, time slice, fuel and year
- $Use_{r,l,f,y}$ is the used primary fuel by the system to satisfy a certain demand of another fuel for each region, time slice, fuel and year

As reflected in equation (3), the objective of integrating RE in the energy system is satisfied through a constraint, imposed by the modeler, ensuring there is a minimum electricity production from RES (Howells et al., 2011).

$$\forall_{\mathrm{r,y}} : TotalREproductionannually_{r,y} \geq \mathrm{REminproductiontarget}_{\mathrm{r,y}} \times REtotaldemandoftargetfuelannually_{r,y} \tag{3}$$

Where:

- r: Region – country modeled
- l: Time slice – fraction representing a period in a year
- y: Year in time horizon
- f: 'Fuel' used or generated by a technology
- $\mathrm{Production}_{\mathrm{r,l,f,y}}$: production of a certain energy to satisfy the demand for each region, time slice, fuel and year (variable)
- $\mathrm{Demand}_{\mathrm{r,l,f,y}}$: demanded fuels for each region, time slice, fuel and year (variable)
- $\mathrm{Use}_{\mathrm{r,l,f,y}}$: used primary fuel by the system to satisfy a certain demand of another fuel for each region, time slice, fuel and year (variable)
- TotalRE production $\mathrm{annually}_{\mathrm{r,y}}$: total energy produced by RES for each year and region (variable)
- $\mathrm{REminproductiontarget}_{\mathrm{r,y}}$: energy systems target to achieve in terms of energy produced by RES for each region and year (parameter)
- $\mathrm{REtotaldemandoftargetfuelannually}_{\mathrm{r,y}}$: total demand that should be fully or partially satisfied by RES for each region r and year y (variable)

This exact optimization through linear programming is based on the following decision variables:

- Activity_rate: the rate of functioning of the technology;
- New_capacity: the new capacities installed annually for each technology which reflects investments;
- Online_capacity: the capacity of each technology connected to the national grid.

In fact, OSeMOSYS identifies the rate of activity (or use) for each technology at each period to produce a competitive price of energy and to satisfy the demand. Installed capacities might be not enough for the needed rate of an activity, therefore OSeMOSYS relies on adding future capacities through investments to enlarge the use of a technology.

In order to refine the analysis of short-term implications due to the penetration of intermittent RE sources, the third type of variables "Online_capacity" is used. It allows considering upward and downward primary and secondary operating reserves. Since the power system represents a combination of conventional and renewable technologies, long-term models have to specify a reserve margin to make sure there are enough capacities and reserves to satisfy the demand. All supply technologies, i.e. production, demand, storage, etc., are subject to reserve conditions. Then, a part of the technical characteristics described above of each technology, maximum reserve contributions and minimum stable generation levels are considered. Therefore, the model figures out the technology that has to be online, i.e. the spinning reserve, to deliver operating reserve convertible instantaneously in terms of energy; and the technology that has to be turned off (Welschet al., 2015), (Gardumi, 2016).

2.2. System reliability integration

Amongst direct load control methods, the system operator may rely in some extreme cases on load shedding in some areas due to the incremental peak or variability of RES and to prevent the collapse of the power system. Taking into consideration cost-benefit analyses in (Neudorfet al., 1995), load shedding might be costly and has negative effects on economic activities since it reduces energy access. Moreover, it is not an efficient technical solution because it causes grid instability (Bhattacharyya, 2011a). Then, system operators tend to satisfy continuously electricity demand and to achieve high rates of system reliability. In this research work, we consider the different definitions of demand-side management actions introduced in (Bhattacharyya, 2011a). Seen otherwise, load shedding could be associated to load management methods and more precisely peak clipping, if it is targeted, well planned, and programmed in advance. Instead of lessening peak load through curtailments in area zones, peak clipping ensures reduction using planned outages with special customers. Several system operators have launched initiatives promoting the non-utilization of electricity in peak periods. This is accomplished through policies, regulatory framework and agreements between the consumer and the system operator. For example, outages tariffs are agreements where the customer accepts electricity outages during a scheduled period in the year and the system operator pays the customer during this period for not providing electricity. The cost paid by the system operator is lower than the cost of satisfying demand during the outage period. This action has been pointed out in the research works about Chile power system in 1992 (Bernstein and Agurto, 1992). Peak clipping method allows decreasing the peak. Therefore, it avoids the system operator from using costly technologies, low efficiency and polluting technologies to meet electricity demand.

Based on the definition that we have adopted of the system reliability in section 1, we could associate a metric to the loss of load. Considering (Prada, 1999), a loss of load will occur whenever the system load exceeds the generating capacity in service. The overall probability that the load demand will not be met is called the Loss-of-

Load Probability (LOLP). Then, LOLP could be used to determine the power system reliability as a parameter to integrate by the modeler for each year and country following equation (4).

$$\forall_{r,l,f,y}\ \mathrm{SystemReliability}_{r,f,y} = \Pr(S \geq D) \text{or } \mathrm{SystemReliability}_{r,f,y} = 1 - LOLP_{r,f,y} \quad (4)$$

Where:

- System $\mathrm{Reliability}_{r,f,y}$: percentage reflecting the system requirements to satisfy the demand (parameter)
- $\mathrm{LOLP}_{r,f,y}$: the ratio of the outages hours during 5 years (intermediate parameter to calculate reliability)
- S: supply of electricity
- D: demand of electricity

Then, we associate peak clipping initiatives to system reliability in OSeMOSYS through the modification of equation (2). Equation (5) links system reliability to the total discounted costs in the energy system structure through the satisfaction of demand.

$$\forall_{r,l,f,y}\ production_{r,l,f,y} \geq demand_{r,l,f,y} \times (SystemReliability_{r,f,y}) + use_{r,l,f,y} \quad (5)$$

To summarize, as an optimization and bottom-up model, OSeMOSYS will satisfy the demand with a supply-oriented focus. The demand to meet is exogenously introduced. Therefore, OSeMOSYS is not a demand-oriented model that could capture load management and energy efficiency actions related to the demand. This work aims to demonstrate that an optimization and bottom-up model could take into consideration demand side characteristics without endogenously modeling the demand. Then, system reliability which is a supply-oriented pseudo variable would represent an energy efficiency measure, for e.g. peak clipping.

2.3. Case study: Tunisia power system

Due to the shortage of fossil fuel based energy, the energy balance in Tunisia has switched from a surplus to a deficit starting from the 2000's, as declared by the Ministry in charge of Energy (Débat national sur l'énergie). Reviews, such as (Dhakouani et al., 2014), have pointed out the negative impacts of this situation on the energy budget, especially that the Government is subsidizing local prices. In fact, the difference between consumer prices and production costs are fulfilled either by the state budget of the power system operator. While production costs are sensitive to high-currency exchange rates and international oil price, in general this decade has been marked by a financial mismatch between the demand and resources.

The lack of financial resources represents a primary barrier in the energy transition, which caused a delay in diversifying resources and reforming the sector. Few years ago, the Government has set a strategy to minimize the dependence on fossil fuels consisting principally of a massive penetration of RES in the power system entitled *"Plan Solaire Tunisien"* (Agence Nationale pour la Maîtrise de l'Energie and 'Plan Solaire Tunisien, 2016). Other measures have been settled such as the regulatory and legislative frameworks in 2015. Based on (Tarifs STEG) electricity generation was satisfied at 98% from natural gas in 2017 using primarily steam turbines, combined cycles and gas turbines. RE technologies are pretty limited and are represented through wind farms and hydropower plants. The national objective aims to achieve 30% of RE integration in electricity generation by 2030 (Agence Nationale pour la Maîtrise de l'Energie and 'Plan Solaire Tunisien, 2016), (Lechtenböhmeret al., 2012), based principally on centralized PV power plants, wind farms, CSP and biomass. Meanwhile, from the 1960's, the Government has worked on electrifying the whole territory; and since the 2000's, Tunisia has achieved over 99% of electrification rate (STEG). Taking into consideration the availability of primary resources, investment planning and electrification rates, the system reliability in Tunisia is well above 99%.

To model the supply chain, technical parameters are identified from previous related research works where the Tunisia power system has been modeled using OSeMOSYS (Dhakouani et al., 2017). Technical-economic characteristics are collected also from (Dhakouani et al., 2017) and updated with real data from the Utility Company (Brand and Missaoui, 2014), (Société Tunisienne de l'Electricité et du Gaz, 2017). Due to changes in public debt ratios since 2010, we have supposed a discount rate of 8% as recommended by the Utility company and declared by the Financial sector (Société Tunisienne de l'Electricité et du Gaz, 2017), (Fassi, 2016). Two scenarios have been assumed as in (Neudorfet al., 1995): i) a BAU scenario reflecting a natural integration of RE in the electricity mix (5% RES by 2030) due, principally, to the decrease of RE technologies prices and natural market penetration, and ii) a RE scenario reflecting the

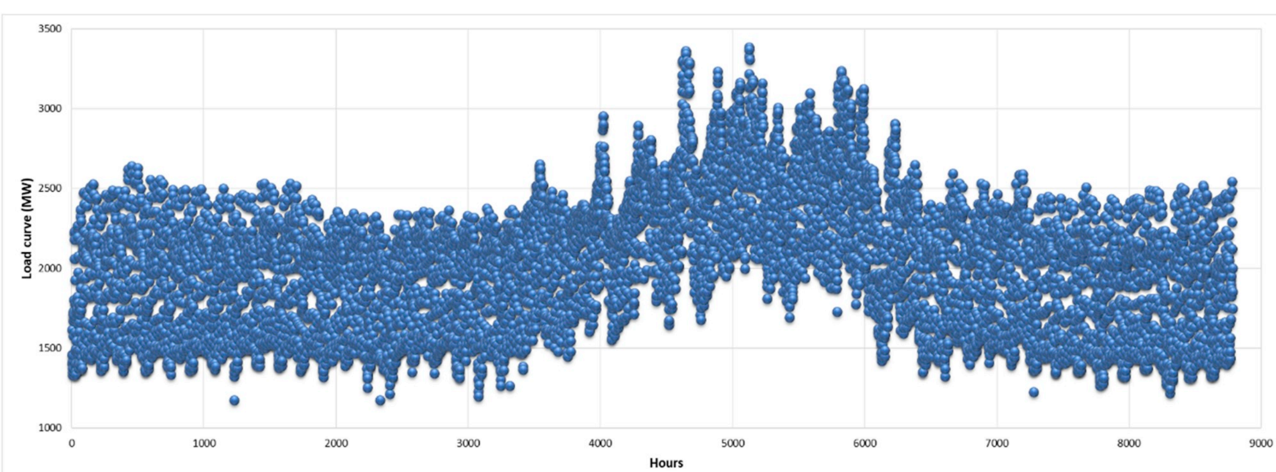

Fig. 1. Annual load curve in 2016.

national objective of 30% of RE in the electricity mix. In brief, both adopted scenarios (BAU and RE) are primarily based on the penetration rate of RES. The simulations will then show renewable technologies shares in the electricity mix (Dhakouani et al., 2017).

The time horizon that is considered is from 2010 to 2030, relating the major changes in the power system to the national objective (Agence Nationale pour la Maîtrise de l'Energie and 'Plan Solaire Tunisien, 2016). Electricity demand starts with 46.85 PJ/year in 2010 followed by real data demand till 2017; for the following years projections are taken from (Lechtenböhmeret al., 2012).

Fig. 1 represents the annual load curve during 2016 (Société Tunisienne de l'Electricité et du Gaz, 2017a). On an annual basis, the electricity peak occurs in summer time primarily due to weather conditions, where the utilization of cooling technologies and appliances increase. The second peak occurs in winter time because of heating, lighting and season-based activities. The troughs are in intermediate seasons, autumn and spring, because of moderate weather conditions and the length of day time. Furthermore, the global warming effects might have slightly decreased the utilization of heating in winter and intermediate seasons. In fact, in Tunisia, an increase in temperatures has been observed in the beginning of the 21st century, varying from one region to another with an average value of 0.35 °C annually. This increase in temperatures has been associated to the impact of climate change (Ministère de l'Environnement et du Développement Durable, 2006). Moreover, researchers in (Mohamed et al., 2007) argue that, starting from 1997, the annual peak demand has switched from the night time to the day time. Considering (Société Tunisienne de l'Electricité et du Gaz, 2017a), it is important to mention that residential electricity demand represents over 30% in 2018, primarily characterized by heating in winter time and cooling in summer time. Thus, we assume that:

- The impact of cooling within the residential sector plays an important role in the increase of the load curve peak; and
- The impact of heating is reflected through the decrease of night peak in winter time and the increase in duration of the annual trough.

Taking into consideration the current tendencies Policy makers and experts argue that policies favoring energy efficiency have also affected the load curve by slightly slowing down the increase of the peaks in the electricity demand profile through the years (Société Tunisienne de l'Electricité et du Gaz, 2016), (Hellal, 2015). However, the increase of the peak still represents the major problem for the system operator of Tunisia. In 2017, it has reached over 4000 MW where all generating capacities were at full load.

Considering the annual load curve, its evolution through the years obeys to an analogy which appeals the peak and the trough growth. In fact, the historical evolution of the demand has shown that the difference between the peak and the trough is reflected by a 5% growth annually, as suggested in Ref. (Hellal, 2015). As shown in Fig. 2, we will adopt the 5% stretch between the annual peak and trough in the demand for the time horizon 2010–2030. Fig. 2 points out the evolution of the demand per time slice during the year. Every year is divided into 4 principal types: intermediate, winter, summer and Ramadhan. Intermediate time slice represents spring and autumn seasons where the weather conditions are mild and the consumption is low. Summer time slice represents summer season and winter time slice represents winter season. The lunar month Ramadhan has been represented through a separate time slice due to the change of the daily profile, i.e. change of behavioral consumption. Each principal time slice is divided into 4 secondary time slices representing the day and night times during working days and days off. For this matter, we have obtained:

- IOD : intermediate open days

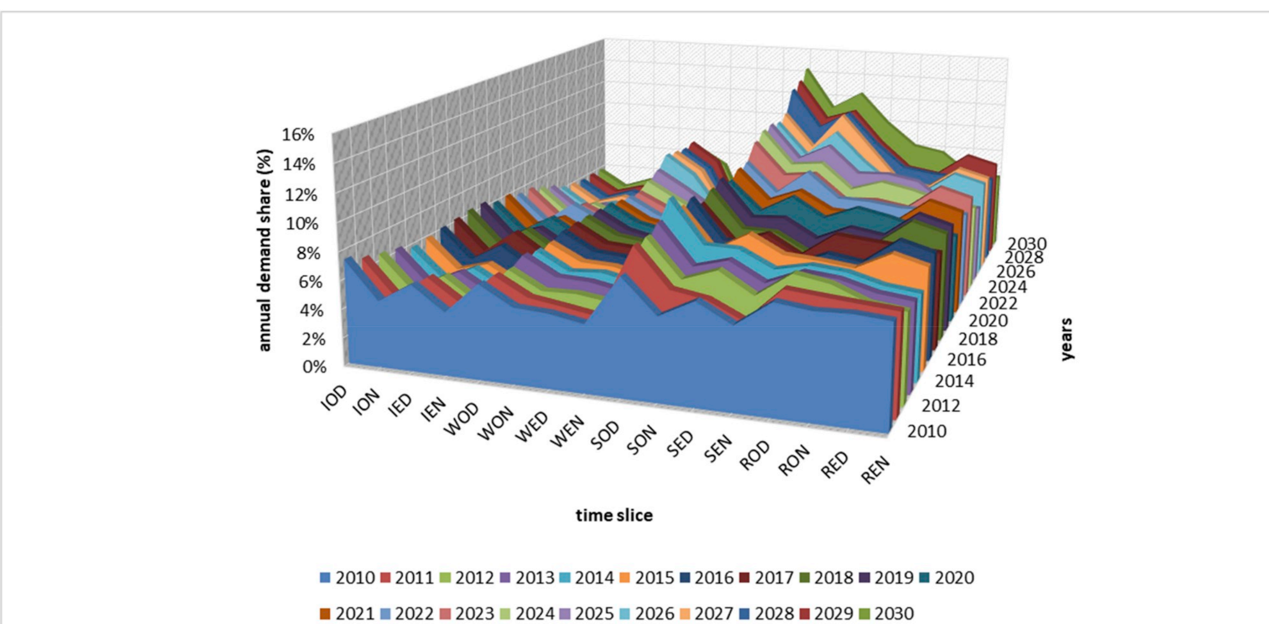

Fig. 2. Estimation of annual demand profiles.

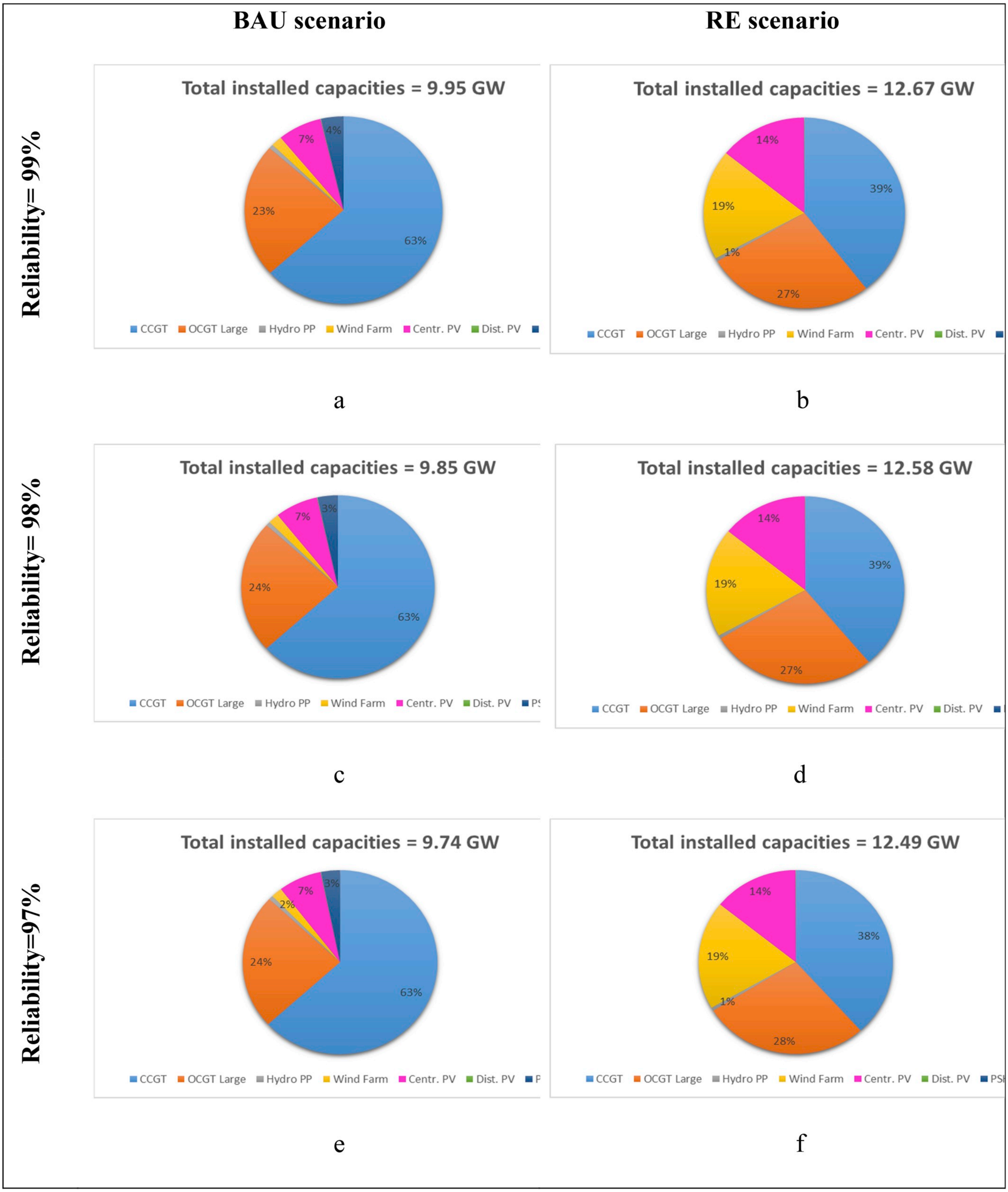

Fig. 3. Impacts of reliability on investments – BAU vs. RE scenario.

Table 1
Reliability impacts on costs – BAU vs. RE scenario.

Costs (Billion USD2010)	Reliability = 0.99	Reliability = 0.98	Reliability = 0.97
BAU	13.59	13.45	13.31
RE	13.83	13.71	13.58

- ION : intermediate open nights
- IED : intermediate off days
- IEN : intermediate off nights
- WOD : winter open days
- WON : winter open nights
- WED : winter off days
- WEN : winter off nights
- SOD : summer open days
- SON : summer open nights
- SED : summer off days
- SEN : summer off nights
- ROD : Ramadhan open days
- RON : Ramadhan open nights
- RED : Ramadhan off days
- REN : Ramadhan off nights

This division has been considered in order to reflect the most important features of the demand annual and daily profile, including peculiarities. In fact, OSeMOSYS allows considering a higher number of time slices which could better reflect the trend of the demand. However, this would increase the simulation time and won't dramatically impact the long-term planning perspectives.

In regard to natural gas prices, an average value of 7.9 USD/PJ for 2010 has been considered, followed by projections taken from (WEO - Energy access database, 2017).

Technologies could be divided into existing and future technologies:

- NG fired Steam PPs, CCGTs, OCGTs (large and small capacities), wind turbines, hydro power plants, decentralized PV (STEG);
- CCGTs, OCGTs (large capacities), wind turbines, decentralized PV, Centralized PV, CSP, PSH (Agence Nationale pour la Maîtrise de l'Energie and 'Plan Solaire Tunisien, 2016).

Further details about the technical-economic inputs of modeled technologies are in Table Annex 1. For each technology, needed technical and economic features are defined and integrated in the model as parameters:

➢ Performance parameters: efficiency, availability factor, capacity factor, residual capacity, operational life time
➢ Cost parameters: capital costs, variable O&M costs, fixed O&M costs, fuel costs
➢ Environmental parameters: penalties per emissions and corresponding fuel specific emission factors.

The operation costs associated to existing technologies have been collected from Refs. (Lechtenböhmeret al., 2012), (Société Tunisienne de l'Electricité et du Gaz, 2016), (Hellal, 2015), while those for about future technologies were assumed from Refs. (IEA, 2009; European Wind Energy Association, 2009; Mehoset al., 2017). All technologies technical parameters, electricity demand and capacity requirements were collected from Refs. (STEG), (Société Tunisienne de l'Electricité et du Gaz, 2017), (Hellal, 2015) except for the ramping characteristics of the power plants, retrieved from Refs. (Meibomet al., 2008; Ela et al., 2011; Aptech, 2012). The environmental parameters related to emissions were taken from Refs. (Société Tunisienne de l'Electricité et du Gaz, 2017; Ministère de l'Environnement, 2010; Tunisian Parliament Ministère de l'Environnement et du Développement Durable, 2005; Tunisian ParliamentMinistère de l'Environnement et du Développement Durable, 2010). In regard to emission penalties, they were estimated based on forecasts presented in (Lechtenböhmeret al., 2012), (Kokorin and Korppoo, 2014), (Yunyi C, 2012).

In fact, the choice of conventional technologies is principally based on the perspectives of the Utility Company on improving energy efficiency through the management of its generation capacities. It has set a strategy for decreasing the consumption of fossil fuels especially that most of the electricity production relies on conventional technologies. Then, energy efficiency is seen more as an upstream activity since it aims at reducing the consumption of primary energy. The recorded values of fossil fuel energy consumption in 2010 and 2011 are due to the decrease in the specific consumption of electricity generating capacities. Indeed, starting from 2014, the prominent efforts, i.e. investments in combined cycle and large open cycle gas turbines, have resulted in an exponential decrease in specific consumption. This has engendered a decrease in utilization of fossil fuels for the production of electricity even though the electricity demand has been increasing in this period. Amongst the activities that have been carried out to improve the energy efficiency in the production cycle:

- Optimization of the geographic allocation of power plants;
- Enhancement of the investment in combined cycle power plants, which are known to have high efficiency rates in comparison with other conventional technologies;
- Establishment of a strategy stopping investments in low energy efficiency technologies, i.e. steam turbines and small open cycle gas turbines (20–30 MW). The Utility company will leave these technologies to decommission and replace them gradually with higher efficiency technologies;
- Investment in higher energy efficiency peak technologies, e.g. class F and above for open cycle gas turbines.

Taking into consideration equation (3), assuming a reliability of 99%, 98% and 97% means that the duration of outages is between 1 and 10 days, 20–30 days and around 40–50 days, respectively, for a period of 5 years.

3. Results and discussions

In this section, we will present supply capacities' shares and costs of the adopted scenarios, i.e. Business As Usual and Renewable Energy, for different assumed reliability system factors.

3.1. Impacts of the system reliability on future investments

Fig. 3 presents installed capacities by 2030 for BAU and RE scenarios at the different system reliability factors considered, i.e. 99%, 98% and 97%.

Comparing Fig. 3 a., Fig. 3 c. and Fig. 3 e., which reflect total needed investments for a BAU scenario in 2030, we can conclude that a decrease of the reliability by 1% entails a decrease in total installed capacities in 2030 by 100 MW. It is important to mention that CCGT's future investments in the BAU scenario didn't change while decreasing system reliability. This is due to the technical characteristics of these technologies classifying it as a base load technology not impacted by the fluctuations of the load curve and principally the change in the peak. However, while decreasing system reliability PSH capacities have decreased and OCGTs have increased. In fact, PSH plays a dual role consisting of smoothing the load curve by decreasing the stretch between peaks and troughs. While varying system reliability, we have primarily decreased the peak. Then, the need of technologies is more oriented towards peak technologies rather than peak and trough technologies. This is explained by an increase of OCGTs in Fig. 3 c. and Fig. 3 e.

Moreover, for the BAU scenario, CCGTs represent the major part in installed capacities satisfying the base load followed by OCGTs satisfying RE intermittencies and load curve fluctuations. RE investments are principally composed of decentralized PV power plants and a limited share of wind technology, around 2%.

Comparing Fig. 3 b., Fig. 3 d. and Fig. 3 f., which reflect total needed investments for a RE scenario targeting 30% of RES penetration by 2030, we can conclude that a decrease of the reliability by 1% entails a decrease in total installed capacities in 2030 by 90 MW. While decreasing system reliability, the shares of CCGTs have decreased and vice versa for OCGTs. The shares of RE technologies are independent of system reliability variation, characterized by 19% of wind turbines and 14% of centralized PV power plants. The lack of investments in decentralized PV is due to its high capital costs; and the lack in hydropower plants is related to the limited potential of barrages in Tunisia.

To conclude, system reliability variation has more important impacts on generating capacities characterized by a dominance of conventional technologies' investments, 100 MW for BAU scenario vs. 90 MW for RE scenario. The shares of technologies didn't change much while varying power system reliability for both scenarios. Namely, 63% and 39% of installed capacities are CCGTs in the BAU and RE scenarios. Major RE technologies in the RE scenario are wind and centralized PV. Targeting an objective of 30% RE by 2030 increases installed capacities by approximately 3 GW comparing to a BAU scenario, independently of the system reliability adopted. This is due to the intermittence of RES, where back-up flexible capacities are needed leading to extra investments.

3.2. Impacts of the system reliability on costs

Table 1 shows that the gains from uninvested and unexploited capacities has generated a decrease in the total cost during the time horizon by 0.14 billion USD 2010 for the BAU scenario and 0.12 billion USD 2010 for the RE scenario. This could be largely exploited in programmed peak clipping initiatives with high energy consumers, especially that the legislative framework is already settled since 2013 (Tarifs STEG). Moreover, we can notice that the total discounted cost of a RE scenario with a reliability of 97% is almost equal to a BAU scenario with a reliability of 99%. In fact, this is explained by the decrease of utilization of conventional technologies during peak periods which are capped with demand side management actions. Then we could conclude that peak clipping initiative could be applied alongside a high penetration of RE to minimize costs and to render renewable technologies investments more profitable.

Even though unprogrammed load shedding avoids blackouts during peak periods; this solution is not socially accepted. Then, we could exploit load shedding in another manner, by better promoting methods of load shedding for high energy consumers through a programmed planning and against outages tariffs. The decrease in system reliability could definitely reflect actions of EE using peak clipping. The gained decrease in costs caused by a decrease in system reliability is due to the decrease in investments of generating capacities and variable costs (reflecting natural gas utilization, operation and maintenance). This could be exploited to achieve RE integration goals.

4. Conclusions and policy implications

In this work, we have linked the power system reliability to energy efficiency actions through modeling the method of peak clipping, which aims to decrease the peak load using scheduled outages. The system reliability factor has been introduced in the core code of a long-term, bottom-up optimization model for energy planning 'OSeMOSYS'. The case study of the Tunisia power system presented insights about energy efficiency actions and their impacts on the integration of renewable energies. We have considered:

- Two supply-based scenarios: a BAU scenario targeting 5% of RE in 2030 and a RE scenario reflecting the national objective of 30% of RE in 2030
- Variability of the system reliability: 99%, 98% and 97%.

The decrease of the reliability of the power system by 1% yielded to a decrease in the costs of RE scenario by 0.12 billion 2010 USD and the costs of the BAU by 0.14 billion 2010 USD; postponing thus future investments and costly operations. It is important to mention that the total cost of the RE scenario at a reliability of 97% is equal to the cost of the BAU scenario at 99%, i.e. 13.58 billion 2010 USD. Achieving 30% of RE in the electricity mix with 97% of power system reliability is as costly as a BAU scenario with 99% reliability. The 2% decrease in system reliability while integrating RE could be addressed through EE actions such as peak clipping. Therefore, decreasing system reliability could be seen as an energy efficiency action boosting the implementation of RES in the power system. Such actions could be integrated alongside Government's RE objectives through outages tariffs or smart meters. Another close-by issue system operators are facing is the load trough, where electricity produced by RES might exceed the demand. In fact, modeling load shifting could present insights on RE capacity investments and show which EE action is more suitable for RE integration in the power system.

Annex 1
Main power generation technologies characteristics

Parameters *		Conventional technology†				Renewable technology					
		fuel fired Steam PP	CCGT	OCGT – Large	OCGT – Small	Wind	Cen PV	Dec PV	CSP	Hydro	PSH
	Units										
Capital cost	USD/kW	220.5	200.9	705.0	428.0	1523.8	3312.8	5114.5	9800	1628.0	2618
Variable costs	M-USD/PJ	0.0	0.4	0.7	0.0	0.1	0.1	0.0	1.24	0.0	3.85
Fixed costs	USD/kW	21.0	12.1	20.0	15.0	12.5	29.0	29.0	67.26	14.4	18.27
Max capacity factor	%	100.00	100.00	100.00	100.00	38	19.5	19.0	47.9	16.29	30.0
Availability factor	%	91.8	94.9	97.0	94.3	97.5	99.0	99.0	96.0	91.0	92.0
Life cycle	Years	37	28	30	45	20	25	25	30	40	40
Efficiency	%	36.36	44.84	31.85	23.87	100.00	100.00	100.00	100.00	100.00	90.00
Efficiency - oil derivatives	%	36.36	44.05	27.81	22.79	-	-	-	-	-	-
Unitary cost of the starts	USD/MW	58.16	53.32	47.86	29.08	-	-	-	90	-	
Ramping rate	MW/min	8	11	15	10	-	-	-	17	-	15
Minimum technical load	% capacity	37.3	41.2	6.6	27.9	0.0	0.0	0.0	1	0.0	6.6
Max prim. Upward reserve	% capacity	20.0	14.0	13.0	12.0	0.0	0.0	0.0	20.0	0.0	13.0
Max prim. Downward reserve	% capacity	20.0	14.0	13.0	12.0	5.0	5.0	5.0	20.0	5.0	13.0
Max sec. upward reserve	% capacity	12.0	6.0	70.0	75.0	0.0	0.0	0.0	60.0	0.0	70.0
Max sec. downward reserve	% capacity	12.0	6.0	70.0	75.0	100.0	100.0	100.0	40.0	100.0	70.0
CO_2 generation	ton/PJ	1.93	1.46	2.38	2.38	0.00	0.00	0.00	0.00	0.00	2.38
CO_2 generation - oil derivatives	ton/PJ	2.67	-	3.18	3.18	0.00	0.00	0.00	0.00	0.00	3.18
NOx generation	10^{-3} ton/PJ	9.36	6.84	10.44	10.44	0.00	0.00	0.00	0.00	0.00	10.44
NOx generation - oil derivatives	10^{-3} ton/PJ	10.11	-	11.88	11.88	0.00	0.00	0.00	0.00	0.00	11.88

* Dashed cells: not applicable, no data available, out of research scope/All data is relative to 2010 – some parameters are evolving during the time horizon.
† Conventional technologies characteristics where primarily extracted from the existing generating capacities. This data might be different from data sheets presented by manufacturers. For example, there are small OCGT that have been implemented since the 1970's and are still functioning few hours a year.

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